EEE303 Tutorial 6

1. A synchronous motor draws 2000 kVA at a power factor of 90% leading. Calculate the approximate power developed by the motor [hp] knowing it has an efficiency of 95%.

2. A 3-phase, 225 r/min Y-connected synchronous motor connected to a 4 kV, 60 Hz line draws a current of 320 A and absorbs 2000 kW. Calculate
   (a) The apparent power supplied to the motor
   (b) The power factor
   (c) The reactive power absorbed
   (d) The number of poles on the rotor
3. Calculate the approximate full-load current of a three-phase, unity power factor synchronous motor rated 3000 hp (2237 kW), 327 r/min, 4000 V, 60 Hz, if it has an efficiency of 97%. Brushless excitation is provided by a 21 kW, 250 V alternator/rectifier, which is mounted on the shaft between the bearing pedestal and the main rotor. What is the value of the field resistance?
4. A synchronous motor has the following parameters, per phase. Draw the phasor diagram and determine:

   $E = 2.4\ \text{kV}; E_o = 3\ \text{kV}$

   $X_s = 2\ \Omega$

   $I = 900\ \text{A}$

   (a) Torque angle
   (b) Active power, per phase
   (c) Power factor of the motor
   (d) Reactive power absorbed (or delivered), per phase
5. From question 4, calculate the line current and the new torque angle if the mechanical load is suddenly removed. Calculate the new reactive power absorbed (or delivered) by the motor, per phase.
6. The 4000 hp, 6.9 kV motor (18 poles) possesses a synchronous reactance of 10 Ω, per phase. The stator is connected in wye, and the motor operates at full-load (4000 hp) with a leading power factor of 0.89. If the efficiency is 97%, calculate the following:
   (a) The apparent power
   (b) The line current
   (c) The value of Eo, per phase
   (d) The mechanical displacement of the poles from their no-load position
   (e) The total reactive power supplied to the electrical system
   (f) The approximate maximum power the motor can develop, without pulling out of step [hp]
7. In question 6, we wish to adjust the power factor to unity. Calculate
   (a) The exciting voltage Eo required, per phase
   (b) The new torque angle